

Draft version

Dynamical systems time series classification

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Keywords: time series; classification; dynamical systems; NeuralODE; inertial measurement unit

1 Introduction

The paper is devoted to a method for time series classification. The *time series* is a finite sequence of variable length of scalar values $x(t_i) = \{x(t_1), \dots, x(t_n)\}$ defined at particular times $t_1, \dots, t_n \in \mathbb{R}$. The *multivariate* time series is a sequence of vector values $\mathbf{x}(t_i) = \{\mathbf{x}(t_1), \dots, \mathbf{x}(t_n)\}$ where values' dimension is $\mathbb{R}^d$. This is equal to having d time series with the same time domain. Everywhere further we assume time series to be multivariate in general. From theoretical point of view it is convenient to assume time series values $\mathbf{x}(t_i)$ to be *random vectors*. Any time series has an associated *class label* $y \in 1 \dots K$ that is also random in general. Here K is the number of classes. A

classifier is a function $\delta(\tilde{\mathbf{x}}(t_i))$ that associates any time series with a class label. We will seek to obtain a "good" classifier in some sense defined later.

There are many methods for time series classification in the literature. *Distance-based* approaches rely on measuring similarity between time series using distance metrics. The metric can be defined using Dynamic Time Warping [1] and the k-Nearest-Neighbors (k-NN) classifier can be used then [2]. *Feature-based* approaches extract finite set of statistical [3, 4] or spectral [5] features from time series for further classification. *Deep learning* approaches uses neural networks to automatically learn time series features from raw train data. Convolutional neural networks [6, 7], recurrent neural networks (RNN) [8] or their combination [9–11] are designed to process sequential data like time series.

In the paper we assume particular model behind time series - *dynamical system*. By that we mean the time-invariant *ordinary differential equation (ODE)* of the form

$$\begin{cases} \frac{d}{dt}\bar{\mathbf{x}}(t) = f(\bar{\mathbf{x}}), \\ \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}_0, \end{cases} \quad (1)$$

where $\bar{\mathbf{x}} \in M$, $f(\bar{\mathbf{x}}) : M \rightarrow M$ is a *vector field*, $\bar{\mathbf{x}}_0$ is a *starting point*, M is an differentiable manifold in general. To find a solution $\bar{\mathbf{x}}(t)$ satisfying (1) on some time interval is to solve an *initial value problem*. Under some regularity conditions on f the solution is unique [12]. The $\bar{\mathbf{x}}(t)$ is called a *phase trajectory*. We associate K different vector fields f_j , $j \in 1, \dots, K$ with each series class. The final series values are noisy phase trajectories

$$\mathbf{x}(t_i) = \bar{\mathbf{x}}(t_i) + \epsilon(t_i), \quad (2)$$

where $\epsilon(t_i)$ are i.i.d. random variables mutual for all classes and any times, having $\mathbb{E}[\epsilon(t_i)] = 0$ and $\mathbb{D}[\epsilon(t_i)] = \sigma^2 < \infty$. Therefore $\mathbb{E}[\mathbf{x}(t_i)] = \bar{\mathbf{x}}(t_i)$ and $\mathbb{D}[\mathbf{x}(t_i)] = \sigma^2$. We would like to obtain a classifier that would mainly rely on the knowledge of the

underlying dynamical systems f_j and would not strongly depend on the randomness from the $\epsilon(t_i)$ so it may be not fully known. We will later show that provided conditions on the noise are enough to obtain an asymptotically accurate classifier. Moreover, we will be able to adapt the classifier to the more general dynamical system described by the *stochastic differential equation (SDE)* [?] of the form

$$\begin{cases} d\bar{\mathbf{x}}(t) = f(\bar{\mathbf{x}})dt + GdW(t), \\ \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}_0, \end{cases} \quad (3)$$

where $W(t)$ is a m -dimensional Brownian motion, G is a $d \times m$ constant diffusion matrix, unique for each class. So in this case phase trajectories are themselves random.

Our approach can be extended for the less direct connection between observed time series and the phase trajectories (2). Imagine that dimensionality of the dynamics (1) is $k \neq d$ and there is an *observation map* $\alpha : M \rightarrow \mathbb{R}^d$ such that

$$\mathbf{x}(t_i) = \alpha(\bar{\mathbf{x}}(t_i)) + \epsilon(t_i). \quad (4)$$

We would like to reconstruct the phase trajectories from the observations to build a classifier. The reconstruction will be based on building an embedding of the manifold M . In general, an embedding of a manifold M is a smooth map $\phi : M \rightarrow N$ into another manifold N that is a diffeomorphism onto its image $\phi(M)$. It follows that phase trajectories $\bar{\mathbf{x}}(t) \subset M$ and $\phi(\bar{\mathbf{x}}(t)) \subset \phi(M)$ are basically identical (more strictly, diffeomorphic). To reconstruct the phase trajectories we need to build $\phi(\bar{\mathbf{x}}(t))$ from the observed time series $\mathbf{x}(t_i)$. One way to archive this is the Whitney Embedding Theorem [22]. If α is a generic map [?] from M to $\mathbb{R}^{2k+1}$ then $\alpha = \phi$ and observed series are the $\phi(\bar{\mathbf{x}}(t))$. In fact, this approach is identical to the (2) but in general requires an excessive $2k+1$ number of time series even when k is rather small. Another

approach is the Takens embedding theorem [23]. It requires only one observed scalar time series $x(t_i)$. If α is a generic map from M to $\mathbb{R}$ then the embedding is given by the map

$$\phi(\bar{\mathbf{x}}) = (\alpha(x), f^{-1}(\alpha(x)), \dots, f^{-m}(\alpha(x))), \quad (5)$$

where $m > 2k$. Substituting $\bar{\mathbf{x}}(t_i)$ gives us $\phi(\bar{\mathbf{x}}(t_i)) = (x(t_i), x(t_{i-1}), \dots, x(t_{i-k}))$ which we call *delay vector* and denote as $\bar{\mathbf{x}}(t_i)$. Illustration of the Takens theorem is on the Fig. 1. Note that Whitney and Takens theorems assume observed series are not noisy. So when we apply them to the noisy $\mathbf{x}(t_i)$ from the (4) the reconstruction $\phi(\bar{\mathbf{x}}(t_i))$ is also noisy and is basically similar to the $\mathbf{x}(t_i)$ from the (2). Further, we would not differentiate reconstructed phase trajectories and the original one.

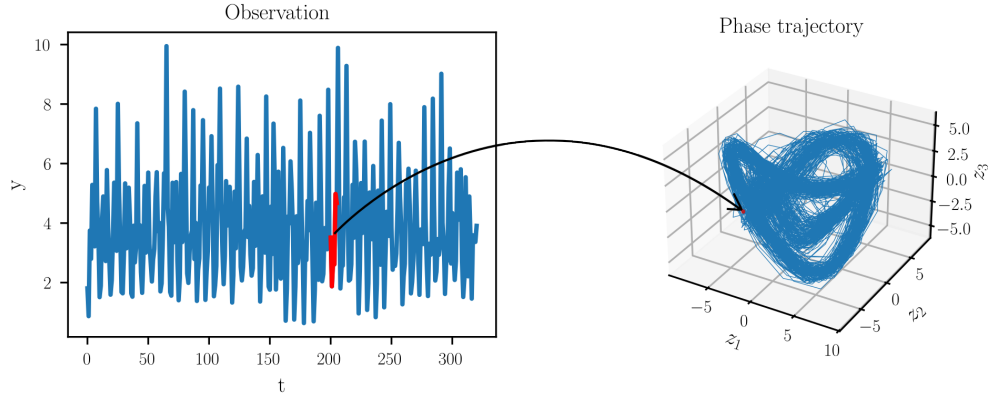


Figure 1 Application of the Takens theorem. On the left are observations $x(t)$. To build a point of the latent phase trajectory $\mathbf{z}(t)$ we must take several lagged observation values (colored in red), e.g. $\mathbf{z}(t_i) = (x(t_i), x(t_{i-1}), x(t_{i-2}))^T$.

Our time series' model can be applied to many real-world classification problems. For example, in medicine an electroencephalogram (EEG) is a direct, macroscopic measure of the brain's electrical activity [13]. The activity is governed by the Maxwell's Equations that can be reduced to the ODEs [14]. The EEG can be used for detecting arrhythmias or other heart conditions [6] as well as to identify seizures and sleep stages from brainwave data [7]. In the weather forecasting the fundamental equations of

atmospheric dynamics can also be reduced to an ODEs system [15]. The classification problems in this field include weather regime classification [16] and climate pattern identification [17]. In human activity recognition (HAR) ODEs are primarily used to model the dynamics of human motion. The core idea is that human movement is a continuous physical process governed by laws of mechanics, which can be described by differential equations [18]. This idea can be applied to the basic or fine-grained activity recognition [19, 20], fall detection [21], and other tasks. To conclude, our model is applicable when it is reasonable to assume that observed time series come from some dynamical system of the form (1) or (3).

Our classifier will rely on the knowledge of the vector fields f_j . But often we do not have them directly but need to build approximations from the data. For this task we will use Neural Ordinary Differential Equations (NODE) [24]. It allows to use gradient optimization when an optimized objective depends on ODE trajectories. There are plenty of papers using NODE for regression tasks. For example, NODE are used in the context of the physics-informed ODE [25, 26] and control learning [27, 28]. However, very few papers analyze NODE in the context of time series classification. Authors of [29] reinterpreted convolutional neural networks in terms of latent ODE dynamics and used it for hyperspectral image classification. Paper [30] solved classification problem in terms of control theory and NODE. Our paper contributes to the NODE applications for the general dynamical systems.

The NODE paper [24] introduced its own popular time series model called latent ODE. It is illustrated on the Fig. 2. The model is probabilistic and in our terms it can be described as

$$\begin{aligned}\bar{\mathbf{x}}_{t_0} &\sim p_0(\bar{\mathbf{x}}_{t_0}), \\ \bar{\mathbf{x}}_{t_1}, \bar{\mathbf{x}}_{t_2}, \dots, \bar{\mathbf{x}}_{t_N} &= \text{ODESolve}(\bar{\mathbf{x}}_{t_0}, f, t_0, \dots, t_N),\end{aligned}$$

$$\text{each } \mathbf{x}_{t_i} \sim p_x(\mathbf{x}_{t_i} | \bar{\mathbf{x}}_{t_i})$$

where $\text{ODESolve}()$ is the solution of the initial value problem. Authors used a variational autoencoder (VAE) [31] to inference the distributions p_0, p_x and the vector field f from the $\mathbf{x}_{t_i}$. While this model is more general and is widely used [32–34], it has several drawbacks compared to ours. First, it has to parameterize f, p_x and the VAE. Hence, more data is needed in general to obtain good approximations of these functions. Our approach may only need to parameterize f . Second, in the latent ODE model a connection between the phase trajectories and the observed time series is given by the general distribution p_x . Thus, the connection is difficult to interpret. In addition, the VAE is needed to reconstruct the $\bar{\mathbf{x}}(t_i)$ from the $\mathbf{x}(t_i)$. Our reconstruction approaches discussed above give explicit and easy ways to obtain $\bar{\mathbf{x}}(t_i)$ from the $\mathbf{x}(t_i)$ and vice versa. Actually, several papers can be found in the literature [35, 36] that use Takens theorem ideas with the NODE to restore latent dynamics. But unfortunately, it is poorly covered for machine learning applications. Our paper aims to close this gap.

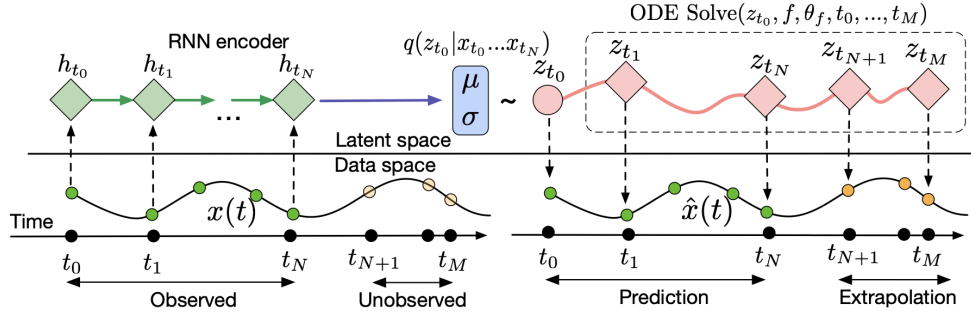


Figure 2 Computational graph of the latent ODE model. Source: reprinted from Chen et al.(2018). Here $z(t_i)$ correspond to the $\bar{\mathbf{x}}(t_i)$

In the next section we will describe a general principal for the time series classification inside our model. We will build an asymptotically optimal classifier under

mentioned properties of the noise ϵ_{t_i} and several additional conditions. These conditions will help with a theoretical prove but may not hold in practice. So we will progressively discard this conditions and adapt classifier for a more general settings. We will also show how adapt the classifier to the SDE case (3). Finally, we will describe the procedure of building vector field approximations from the data. Then we will apply our method to three classification tasks for emprirical validation. The first will be the classification of several known chaotic [?] dynamical systems. The second will be classification of the human motions from the Inertial Measurement Unit (IMU) sensors. The last one will be classification of heart condition from the ECG recordings.

2 Asymptotically optimal classifier

3 Classification in general case

4 Empirical study

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